



Physics-enhanced performance prediction and intelligent design for wide-operating-range turbine blades

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ARTICLE INFO

Communicated by: Dr Pooneh Aref

Keywords:

Physics-enhanced
Similarity principle
Turbine blade
Intelligent rapid design

ABSTRACT

To address the prohibitive computational cost of traditional performance evaluation and the limited generalization of conventional data-driven models in turbine blade design, a physics-enhanced performance prediction model for wide-operating-range turbine blades is proposed. Integrating physics-guided feature engineering and a knowledge-embedded architecture, this method first leverages fluid dynamic similarity principles to non-dimensionalize geometric and aerodynamic inputs, establishing a unified representation across varying scales. Subsequently, a ResUNet-based architecture is constructed to map these inputs to surface isentropic Mach number distributions, serving as intermediate physical constraints, before estimating total pressure loss coefficients. Validated against a diverse dataset of multiple blade profiles, the model captures complex flow structures with high accuracy, achieving a Mean Relative Error (MRE) of 1.16% for flow fields and a Mean Absolute Error (MAE) of 0.26% for loss coefficients. Furthermore, an intelligent inverse/ forward design framework integrating this predictor is established. This framework enables efficient forward and inverse design, demonstrating exceptional stability and accuracy under multi-point operating conditions. These results underscore the method's engineering value for the rapid, robust design of complex turbomachinery.

1. Introduction

Turbines serve as core components in power propulsion systems such as aero-engines and gas turbines, where the aerodynamic performance of their blade profiles is a determinant of overall engine efficiency. Traditional turbine design workflows rely on performance evaluation methods like computational fluid dynamics (CFD) and experiments. However, these approaches face significant bottlenecks. On one hand, the CFD simulations are computationally expensive, and experimental tests incur prohibitive costs. On the other hand, the CFD-based iterative design process is excessively lengthy, making it difficult to meet modern engineering demands for rapid design. Furthermore, traditional design processes depend heavily on engineering experience. The initial selection and optimization direction of blade parameters are often constrained by the designer's subjective knowledge. This reliance not only leads to local optima but also makes it challenging to comprehensively cover diverse performance requirements.

With the rapid advancement of artificial intelligence, data-driven models based on deep learning have gained widespread adoption in the field of turbomachinery flow prediction due to their superior

nonlinear fitting capabilities [1–5]. Early research predominantly utilized foundational architectures such as multi-layer perceptron (MLP) [6] and convolutional neural network (CNN) [7], achieving fast and accurate predictions of flow fields and performance parameters, thereby reducing single-point evaluation times to the millisecond level. For instance, Sekar et al. [8] combined CNNs with MLPs to efficiently predict surface pressure and velocity fields for airfoils under incompressible flow conditions, using geometry, Reynolds number, and angle of attack as inputs. To address the difficulty of obtaining variable operating characteristics during the design phase, Wang et al. [9] integrated Generative Adversarial Networks (GANs) with dual CNNs, establishing a rapid mapping from blade profiles to flow reconstruction and performance evaluation, which reduced multi-condition analysis time to seconds. Ji et al. [10] employed Proper Orthogonal Decomposition (POD) to implicitly encode flow field information, accurately predicting centrifugal impeller performance curves via neural networks. Additionally, addressing secondary flow losses in low-aspect-ratio cascades, Du et al. [11] proposed a Series Convolutional Neural Network (SCNN), which achieved high-precision prediction with limited training data, demonstrating significant potential for end-wall optimization.

Despite demonstrating high accuracy under specific conditions,

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<https://doi.org/10.1016/j.ast.2026.112351>

Received 28 January 2026; Received in revised form 23 March 2026; Accepted 10 April 2026

Available online 12 April 2026

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Nomenclature*Variables*

C	Chord length (mm)
C_x	Axial chord length (mm)
Ma	Isentropic Mach number
Ma_1^{is}	Exit isentropic Mach number
P^*	Total pressure (Pa)
P_{ss}	Suction side control points
P_{ps}	Pressure side control points
Re	Reynolds number
R_{le}	Leading – edge radius (mm)
T^*	Total temperature (K)
k	Correlation coefficient
t	Pitch (mm)
x	Axial coordinates (mm)
y	Polar coordinates (mm)

Greek symbols

α	Inlet incidence angle (°)
β	Inlet flow angle (°)
β_1	Inlet geometric angle (°)
β_2	Effective exit angle (°)
θ	Exit deflection angle (°)
γ	Tangential chord angle (°)
ε_1	Inlet wedge angle (°)
ε_2	Exit wedge angle (°)
δ	Total pressure loss coefficient

Abbreviation

AE	Absolute error
CFD	Computational fluid dynamics
MRE	Mean relative error
MAE	Mean absolute error
P-MLP	Physics-enhanced MLP
P-Unet	Physics-enhanced Unet
P-ResUNet	Physics-enhanced ResUNet

current data-driven models face significant limitations when handling complex turbine flow prediction tasks [12,13]. First, these models rely heavily on high-quality labeled data. In the field of turbomachinery, acquiring high-fidelity simulation or experimental data is extremely costly, and sample scarcity severely constrains adequate model training. Second, pure black-box mappings often lack underlying physical constraints, potentially producing results that violate fundamental fluid dynamic conservation laws, making prediction schemes difficult to validate for engineering feasibility. The most critical bottleneck, however, is limited generalization capability; that is, prediction performance degrades significantly when testing samples deviate from the training distribution. Specifically, turbines consist of multiple blade rows with varying geometric dimensions (e.g., pitch, axial chord) and significantly different inlet/outlet boundary conditions. Yet, existing deep learning models mostly focus on specific cascades, with training samples confined to narrow fluctuations around baseline geometries and conditions. Such models, established for specific distributions, suffer from poor numerical stability and accuracy when facing large numerical deviations across different blade rows. Therefore, establishing a robust and universal flow prediction model that can handle significant variations in geometry and boundary conditions remains a key challenge.

To overcome these generalization bottlenecks, the concept of physics-enhanced machine learning has emerged and attracted widespread attention [14–17]. The core concept involves integrating a prior physical laws into data-driven architectures by guiding feature selection, optimizing network structures, or introducing physical criteria as regularization terms in loss functions. This forces the model to adhere to fluid dynamic constraints while fitting data, thereby enhancing accuracy and applicability. Scholars have explored various embedding methods aimed at fusing basic principles, such as similarity theory and continuity equations, to improve robustness using multi-source physical knowledge. In terms of feature selection, Chen et al. [18] utilized deep neural networks to establish physical mappings between key dimensionless parameters (e.g., Mach number, Reynolds number) and optimal loading, subsequently conducting turbine blade inverse design using multi-output Gaussian processes. Regarding network architecture, Li et al. [19] combined simple sweep theory with 2D airfoil data to effectively mitigate the scarcity of 3D blade samples, improving 3D flow prediction accuracy. Bruni et al. [20] constrained intermediate variables via physical equations, significantly reducing data dependence and enhancing extrapolation capabilities. Similarly, research by Pongetti et al. [21] demonstrated that integrating flow field prior knowledge effectively improves performance prediction accuracy for Rotor-37.

Nevertheless, existing physics-enhanced studies exhibit limitations. First, most models focus on specific turbine profiles, lacking sufficient universality for cross-profile and wide-operating-range tasks. Second, even with physical constraints, numerical stability remains weak for unseen samples that deviate significantly from the training distribution. For instance, while Li et al. [22] utilized graph neural networks to improve predictions for low-aspect-ratio turbines, the generalization performance could not escape dependence on the initial training distribution. Consequently, constructing a model that integrates physical knowledge and possesses broad adaptability remains a critical breakthrough direction for realizing intelligent design of turbine blades.

To address the insufficient generalization of conventional models under varying geometric dimensions and wide operating conditions, this paper proposes a physics-enhanced performance prediction model by fusing fluid dynamic similarity theory with deep learning architectures. It is important to clarify that the term physics-enhanced herein denotes a synergy of physics-guided feature engineering and a knowledge-embedded architecture. Distinct from Physics-Informed Neural Networks (PINNs) that minimize PDE residuals, the proposed method respects physical laws through rigorous input constraints based on similarity principles and intrinsic data relationships within the network structure. The main innovations are summarized as follows:

- (1) A unified input representation based on physics-guided feature engineering. By rigorously applying geometric and aerodynamic similarity principles (e.g., non-dimensionalizing coordinates and introducing isentropic Mach numbers), a unified representation paradigm is established. This approach effectively decouples the model from physical scales and reduces the dimensionality of the design space, ensuring robust adaptability across varying blade magnitudes and operating conditions.
- (2) A knowledge-embedded architecture utilizing intermediate physical constraints. Unlike traditional end-to-end methods that directly predict scalar performance, this study introduces the surface flow field distribution as a critical intermediate physical constraint. Implemented via the P-ResUNet architecture, this mechanism guides the model to learn the intrinsic mappings between inputs and complex flow structures, thereby significantly enhancing generalization stability on unseen samples.
- (3) An intelligent design framework balancing high precision and efficiency. By integrating the physics-enhanced predictor with optimization algorithms, a rapid design workflow is constructed. This framework achieves millisecond-level evaluation speeds

while maintaining CFD-level accuracy, effectively overcoming efficiency bottlenecks and enabling robust design under wide-ranging, multi-point operating conditions.

This paper is organized as follows: Section II details the physics-enhanced data-driven model architecture and the construction strategy for a wide-range dataset covering multiple turbine profiles. Section III validates the wide-range generalization capability of the proposed model by comparing prediction accuracy against different baseline models and using multiple test cases. Section IV demonstrates the intelligent forward/inverse design workflow based on the prediction model through typical application examples, verifying its feasibility and efficiency in practical design tasks. Conclusions are drawn in Section V.

2. Performance prediction modeling methodology

To achieve high-precision flow prediction and efficient turbine blade design across wide operating ranges, a physics-enhanced data-driven method is proposed [23]. First, similarity principles are utilized to non-dimensionalize input features, effectively reducing the design space dimensionality. Second, a tailored deep neural network is constructed to capture complex flow features. Finally, an intelligent workflow is established based on this highly generalizable model to realize both forward optimization and inverse blade design.

2.1. Architecture of physics-enhanced data-driven model

2.1.1. Feature non-dimensionalization based on similarity principles

Traditional aerodynamic prediction models often suffer from a sharp decline in accuracy when facing significant variations in geometric parameters (e.g., chord length, pitch) or boundary conditions. To address this bottleneck, a modeling method enhanced by prior physical knowledge is proposed. Guided by fluid dynamic similarity principles, this method non-dimensionalizes the geometric and aerodynamic

parameters. Consequently, a prediction framework with wide-range adaptability is established, as shown in Fig. 1.

Regarding geometric representation, a unified representation for turbine blades of different scales is achieved based on geometric similarity [24]. Two-dimensional blade coordinates and pitch are normalized using the chord length C as a reference. This approach eliminates the model's dependence on actual physical dimensions and reduces the dimensionality of the design space. Specifically, the relative pitch t/C is extracted as a key geometric input feature to represent cascade solidity [25].

For operating condition representation, dimensionless criterion parameters are selected based on kinematic similarity. Given the inlet total temperature T_0^* , the exit isentropic Mach number Ma_1^{is} is introduced to establish a mapping between inlet total pressure P_0^* and outlet static pressure P_1 , thereby characterizing flow compressibility. Simultaneously, the inlet incidence angle α is adopted to reflect the directionality of the incoming flow. Notably, typical turbine blades usually operate within the self-modeling region of the Reynolds number $Re > 5 \times 10^4$, where viscous effects on the flow field are relatively minor [26]. Furthermore, accuracy verification indicates that including Re as an input feature degrades prediction performance; therefore, it is excluded from the input feature space in this model.

Regarding output features, the surface isentropic Mach number distribution Ma is selected to represent loading distribution based on dynamic similarity. Additionally, the total pressure loss coefficient δ is employed as a dimensionless indicator for aerodynamic loss.

2.1.2. The framework of physical information correlation

Traditional prediction models often employ end-to-end direct mapping, which frequently fails to fully exploit the rich spatial information of physical fields contained within the dataset. To further enhance model robustness, a performance prediction framework correlating loading distribution is proposed, as illustrated in Fig. 2.

Building on dimensionless input features, this framework leverages

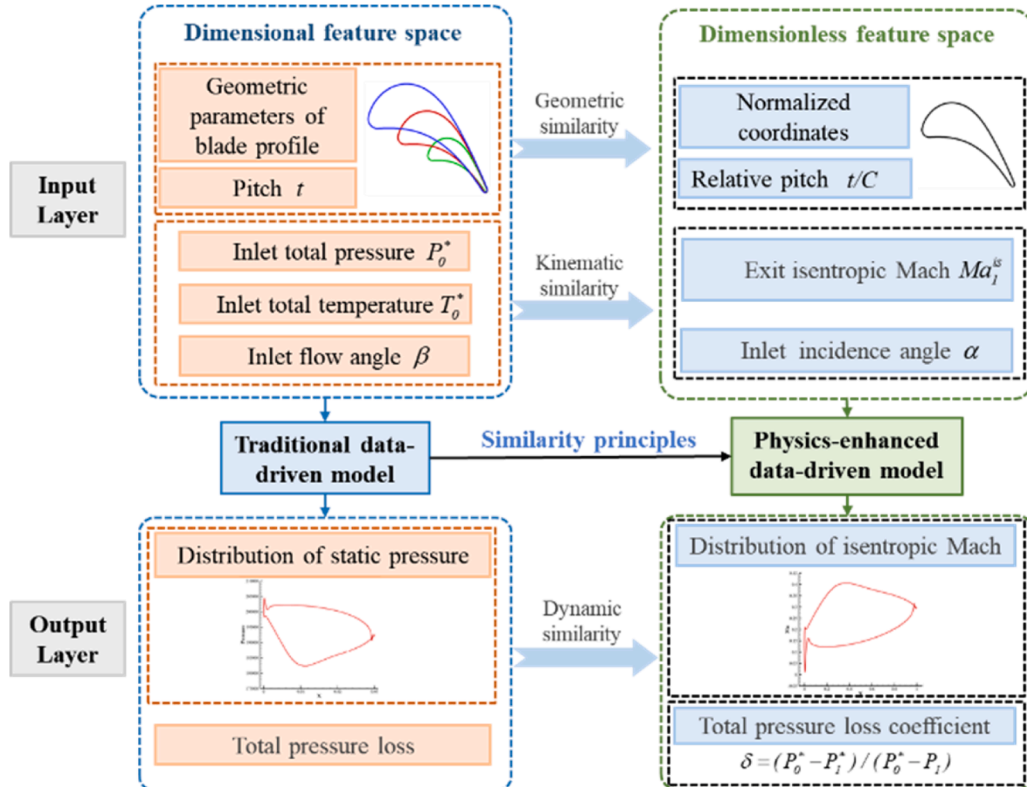


Fig. 1. Framework of feature selection based on similarity principles.

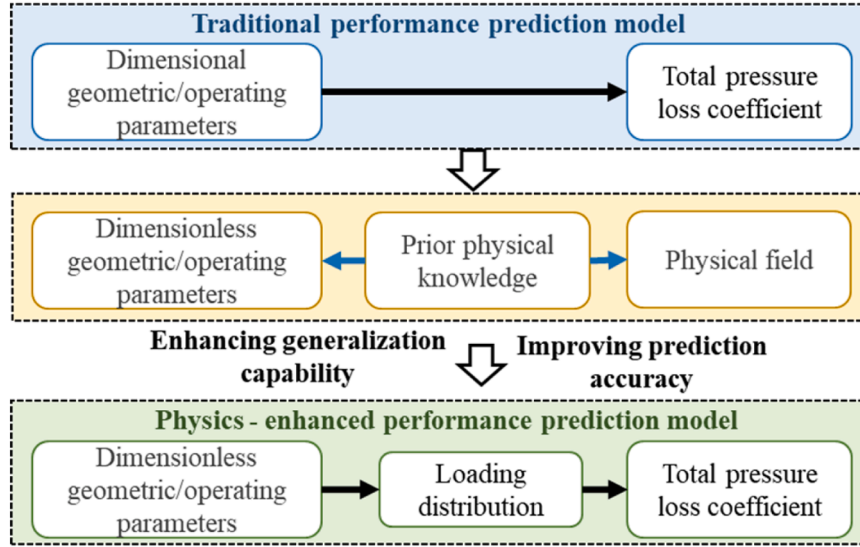


Fig. 2. Schematic of the physics-enhanced modelling approach.

the flexibility of the multi-input multi-output architecture of deep neural networks. During training, an intrinsic physical correlation is established among input features, loading distribution, and performance parameters. This constraint compels the network to adhere to fluid dynamic evolution laws when approximating performance data. This correlated prediction mechanism not only improves generalization capabilities under cross-condition scenarios but also significantly enhances the accuracy of capturing complex flow structures, such as shock waves and separation zones. This lays a solid foundation for the subsequent intelligent forward and inverse design processes.

2.2. The framework of deep neural network

To accurately capture the complex nonlinear mapping between turbine blade parameters and aerodynamic performance, this study adopts a deep neural network architecture with robust feature extraction capabilities. This section first introduces the foundational U-Net framework. Subsequently, the ResUNet architecture, formed by integrating U-Net with ResNet, is described. Finally, the structure of the residual module, serving as the core unit, is analyzed in detail.

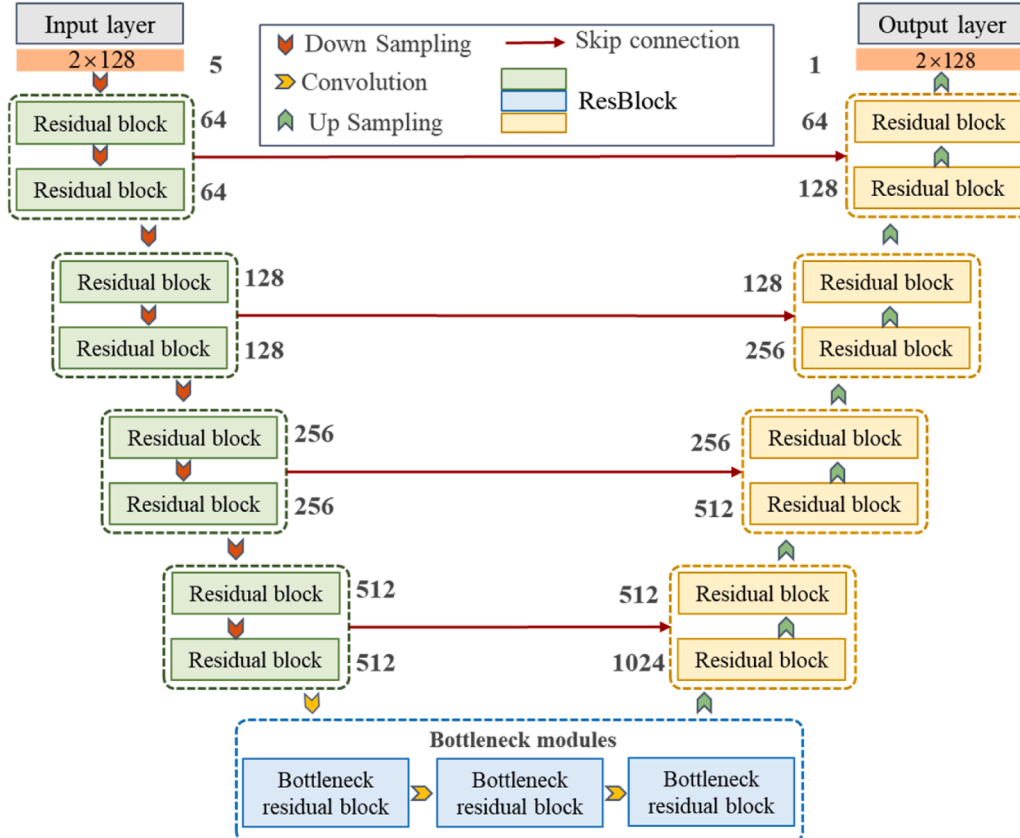


Fig. 3. The architecture of ResUNet.

2.2.1. U-Net

U-Net was originally renowned for its performance in medical image segmentation, attributed to its symmetric encoder-decoder structure [27]. Its capability for precise localization renders it equally suitable for predicting surface flow fields of turbine blades [28,29]. The overall network structure resembles the letter “U” comprising an encoding path on the left and a decoding path on the right, as illustrated in Fig. 3.

(1) Encoder

This path is responsible for extracting semantic information from input feature maps layer by layer. In this study, the encoder extracts geometric features of turbine coordinates and aerodynamic operating parameters through successive convolution and downsampling operations. It typically consists of 4 to 5 convolution modules. Each module includes convolutional layers, max-pooling layers, and activation function layers, designed to deepen feature representation while performing spatial downsampling.

(2) Decoder

This path is responsible for progressively restoring deep, low-resolution features to their original dimensions. Consequently, it establishes the nonlinear mapping between geometric/aerodynamic information and the flow field distribution.

(3) Skip connection

This is a defining characteristic of U-Net. It directly transfers high-resolution local details from the encoding path—such as key geometric features of the leading and trailing edges—to the decoding path. These features are concatenated with upsampled deep features along the channel dimension. This mechanism effectively compensates for spatial information loss caused by downsampling, enabling the model to capture both global flow patterns and local flow details.

2.2.2. ResUNet

Although U-Net possesses distinct advantages in feature extraction, purely convolutional architectures are prone to gradient vanishing or performance degradation as network depth increases [30]. To address this, the concept of residual learning is introduced to construct the ResUNet architecture [31]. ResUNet inherits the symmetric skeleton and skip connections of U-Net. The primary improvement lies in replacing the original standard convolution blocks with residual blocks. This modification yields significant advantages. First, the shortcut connections within residual blocks significantly mitigate the difficulty of training deep networks, enabling the construction of deeper architectures to learn complex flow features across varying operating conditions. Second, the features transmitted via skip connections undergo residual optimization, containing more robust multi-scale information, which significantly enhances prediction accuracy and stability [32].

2.2.3. Residual modules

Serving as the cornerstone of ResUNet, two types of residual blocks are adopted in this study to balance computational efficiency and accuracy requirements, as shown in Fig. 4.

(1) Standard residual block

This module consists of two 3×1 convolutional layers, a Group Normalization layer [33], and ReLU activation functions. The core function of this module is to learn the residual mapping. This design ensures that even in extremely deep networks, the module can degenerate into an identity mapping, thereby preventing performance

degradation.

(2) Bottleneck residual block

To control computational costs while increasing network depth and enhancing nonlinear fitting capability, a bottleneck design is introduced. This structure utilizes a combination of 1×1 convolution - 3×1 convolution - 1×1 convolution, which substantially reduces the number of parameters. When channel dimensions do not match, a 1×1 projection convolution is employed to achieve dimensional alignment.

By flexibly stacking the aforementioned residual units, the constructed ResUNet model achieves high-precision modeling of complex physical quantities in turbine blade flow fields while maintaining training efficiency. This provides a reliable performance evaluator for the subsequent intelligent forward/inverse design workflows.

2.3. Intelligent design method based on neural networks

To realize efficient and intelligent design of turbine blades, an integrated intelligent design framework is constructed. With the physics-enhanced flow prediction model serving as the core for rapid performance evaluation, the framework integrates loading-oriented inverse design and performance-oriented forward optimization through flexible configuration of design objective functions.

As illustrated in Fig. 5(a), the framework integrates the high-precision prediction model with an efficient optimization algorithm [34], establishing a rapid design pathway from design targets to the final blade profile. The workflow adopts an “inverse-then-forward” strategy. First, inverse design is conducted based on specified boundary conditions and target loading distributions to rapidly generate a baseline profile that meets loading requirements. Subsequently, using this baseline profile as the initial solution, forward design is executed to minimize the total pressure loss coefficient. This step performs fine-grained optimization within the local solution space, ultimately yielding an optimal profile that balances ideal loading characteristics with superior aerodynamic performance. This strategy effectively overcomes the traditional reliance on empirical experience for initial profile selection, significantly enhancing the robustness and quality of the design results.

To further elucidate the algorithmic, Fig. 5(b) and Fig. 5(c) present detailed flowcharts for the inverse and forward design methods, respectively. Both processes comprise the following three key stages:

(1) Problem definition and parameter configuration

First, the optimization objective is defined according to the specific task. In inverse design mode, the target loading distribution is imported; this distribution is typically extracted from optimal profiles in historical databases and uniformly resampled to standardize the spatial discretization. In forward design mode, performance optimization metrics are directly established, such as minimizing the total pressure loss coefficient. Subsequently, fixed aerodynamic boundary conditions and geometric constraints are configured, and the search space for geometric design variables is delimited. Finally, the pre-trained flow prediction model is loaded to serve as the tool for rapid performance evaluation.

(2) Iterative optimization and performance evaluation

This framework employs the Differential Evolution (DE) algorithm as the driving engine. During iterations, the algorithm generates geometric design variables for candidate profiles and maps them into standard input tensors for the prediction model. In inverse design mode, the model outputs the predicted loading distribution of the candidate

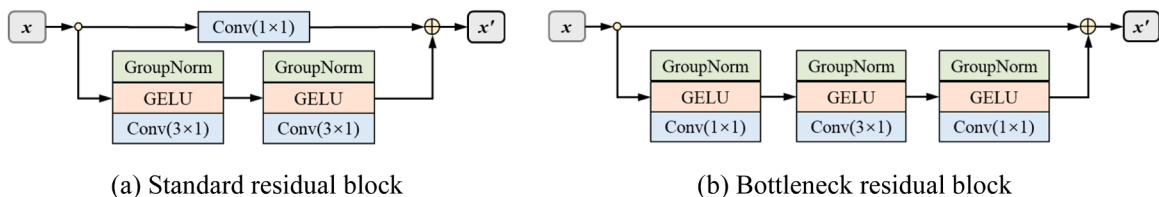
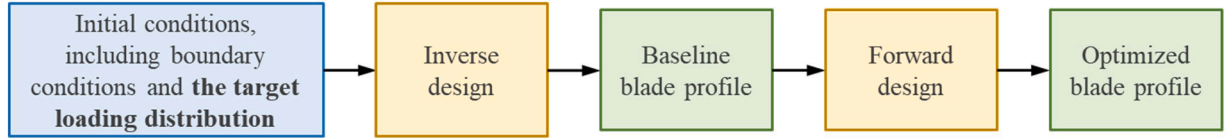
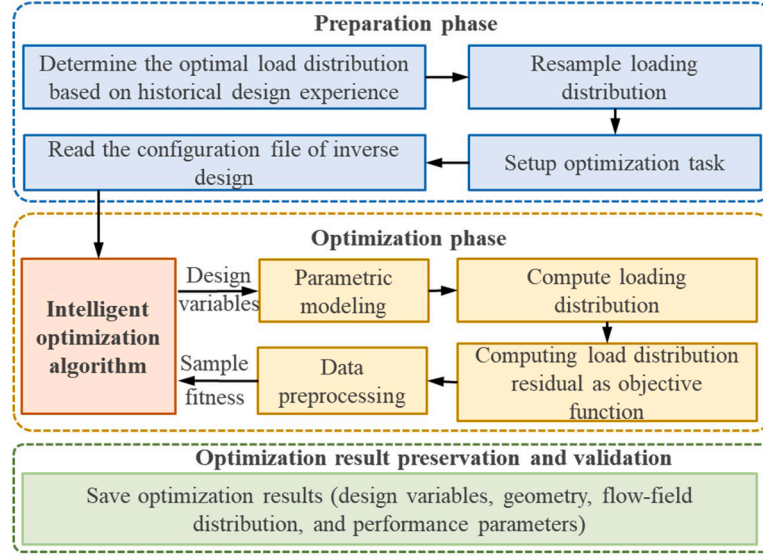


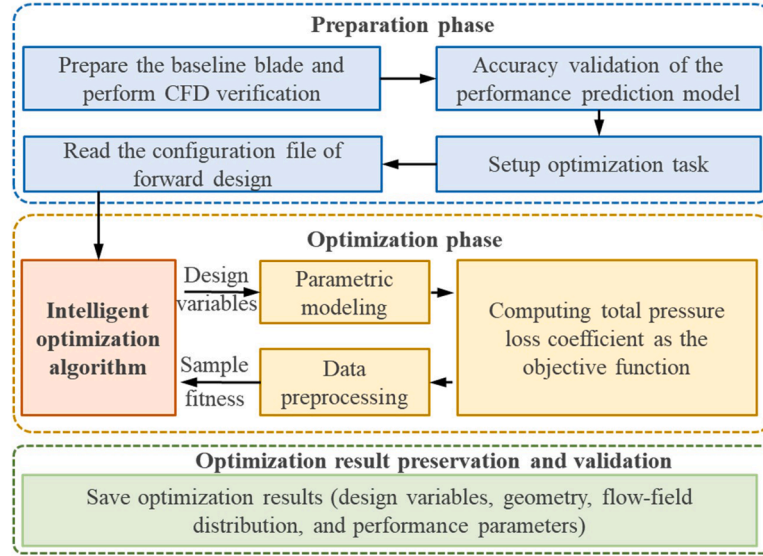
Fig. 4. The structure of a residual block.



(a) Intelligent design method



(b) Inverse design



(c) Forward design

Fig. 5. Intelligent design method based on data-driven models.

profile, and the root mean square error (RMSE) relative to the target distribution is calculated as the fitness metric. In forward design mode, the model directly predicts key performance parameters, utilizing indicators such as the total pressure loss coefficient as the objective function. The optimization algorithm updates the population based on the feedback fitness values, driving the search process toward the global optimal solution.

(3) Result generation and output

The optimization process terminates when convergence criteria are met or the preset maximum number of generations is reached. The system extracts the optimal design variables, the corresponding blade profile coordinates, the predicted flow field distribution contours, and

all key aerodynamic performance indicators.

The advantages of this framework lie in its architectural flexibility and computational efficiency. A unified architecture adapts to diverse design requirements. Furthermore, the deep learning-based evaluation approach replaces time-consuming CFD simulations, reducing the duration of a single performance assessment from minutes to milliseconds. This order-of-magnitude improvement in efficiency empowers the capability to search across extensive design spaces, significantly accelerating both the preliminary design iteration and the global optimization process for turbine blades.

3. Numerical simulation setting and dataset configuration

Constructing a high-quality and diverse dataset is fundamental to training data-driven models with wide operating ranges. This section details three key aspects of dataset construction: First, a multi-type blade geometry database is established to provide sufficient geometric samples. Second, the numerical simulation method is validated to ensure data accuracy and reliability. Finally, a structured dataset covering a wide range of operating conditions is generated.

3.1. Construction of the blade database

To construct a prediction model suitable for wide-range turbine blade design, a geometric dataset containing multiple blade types is established in this study. The baseline database encompasses various industrially representative profiles. These include two typical power turbine blades, the GE-E3 high-pressure turbine blade [35], and the PW-E3 high/low-pressure turbine blades [36]. Significant differences exist among these profiles regarding geometric scale, loading characteristics, and application scenarios. This diversity provides the necessary geometric variety for the model. To achieve unified feature representation and efficient sampling, an improved parameterization method is adopted [37]. Key parameters such as axial chord length C_x , inlet geometric angle β_1 , inlet wedge angle ϵ_1 , tangential chord angle γ , exit deflection angle θ , and effective exit angle β_2 are selected. These parameters map complex blade contours into quantifiable geometric parameter vectors, as shown in Fig. 6.

Based on these baseline profiles, the Latin Hypercube Sampling (LHS) method is employed to expand the multi-dimensional parameter space. Consequently, a database containing 20,000 blade samples is generated, significantly expanding the geometric design space. To intuitively reveal the geometric feature distribution within the database, the t-SNE dimensionality reduction method [38] is utilized for

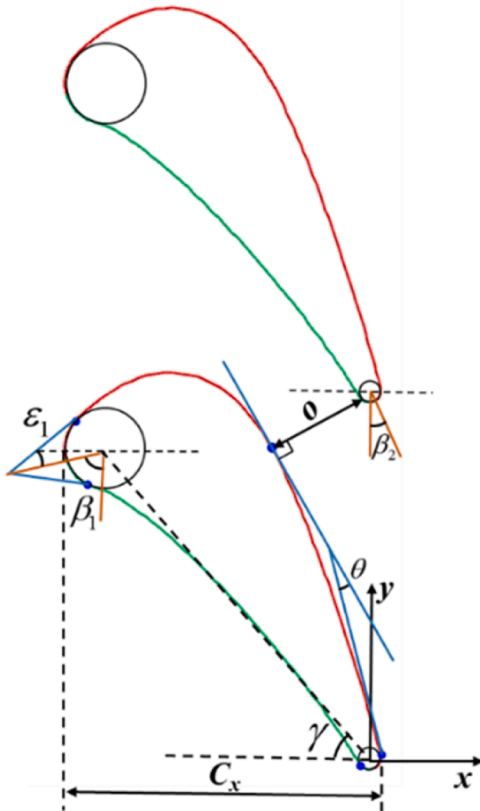


Fig. 6. Parametrization methods for turbine blades.

visualization, as illustrated in Fig. 7. In the figure, each scatter point represents a blade sample. The Euclidean distance between points reflects the similarity of geometric features. Different colored clusters correspond to different baseline turbine blade types. From the perspective of feature topology, different blade types exhibit relatively independent clustering in the low-dimensional space. Simultaneously, they demonstrate good continuity and transition at the cluster boundaries. This distribution confirms the extensive feature coverage of the database. It lays a solid geometric data foundation for training a prediction model with wide-range adaptability. Furthermore, visual analysis indicates a specific trend. Along the direction of increasing Dimension 1 (indicated by the red dashed line), the inlet geometric angle of the turbine blades shows a monotonically increasing evolution.

3.2. Validation of numerical simulation methods

To ensure the fidelity of the flow fields and performance parameters, an automated CFD simulation workflow was established using the ANSYS CFX platform. The numerical model was constructed for a two-dimensional linear cascade, as illustrated in Fig. 8. The $k-w$ Shear Stress Transport (SST) turbulence model was selected to accurately resolve near-wall flow features [39]. To precisely capture the complex boundary layer flow, the near-wall mesh was refined, and the first-layer grid height was strictly controlled to ensure a dimensionless wall distance of $y^+ = 1$, thereby satisfying the requirements of the SST turbulence model for resolving the viscous sublayer. Furthermore, to minimize numerical dissipation and improve numerical accuracy, the High Resolution scheme was adopted for both the Advection Scheme and Turbulence Numerics in the solver. The convergence criterion for the steady-state calculations was set such that the root-mean-square (RMS) residuals of all governing equations dropped below 10^{-6} , guaranteeing the high fidelity of the predicted flow fields and aerodynamic performance. Additionally, the computational domain was extended sufficiently upstream and downstream to ensure fully developed flow conditions. Regarding aerodynamic boundary conditions, total pressure, total temperature, and flow angle were imposed at the inlet, while static pressure was specified at the outlet.

To ensure the reliability of the numerical results, methodological validation was conducted. A grid independence study demonstrated that key aerodynamic indicators, such as the total pressure loss coefficient, achieved convergence when the grid count reached approximately 90,000. This configuration represents an optimal balance between computational accuracy and efficiency, as shown in Fig. 9.

Furthermore, CFD results were validated against experimental data under three typical incidence angles: -20° , 0° , and 10° , as depicted in Fig. 10. The comparison reveals that, the predicted surface isentropic Mach number distributions exhibit high agreement with experimental measurements, particularly in the suction-side acceleration region. These results confirm that the adopted numerical method can accurately capture loading distribution characteristics and detailed flow structures. Consequently, it provides a solid foundation for generating a high-quality and high-fidelity flow field dataset for training.

3.3. Dataset preparation

In the dataset preparation phase, sampling boundaries for key geometric and aerodynamic input features are explicitly defined. This ensures the model possesses operational adaptability. Specifically, the range for relative pitch t/C is set to $[0.3, 1.0]$. The exit isentropic Mach number Ma_t^s covers the interval $[0.2, 1.2]$, while the inlet incidence angle α spans from $[-30^\circ, 15^\circ]$. These parameter ranges achieve broad coverage of geometric solidity, inlet flow direction, and outlet compressibility. Consequently, the dataset comprehensively encompasses diverse typical states, ranging from high-load vanes in high-pressure turbines to conventional blades in low-to-medium pressure

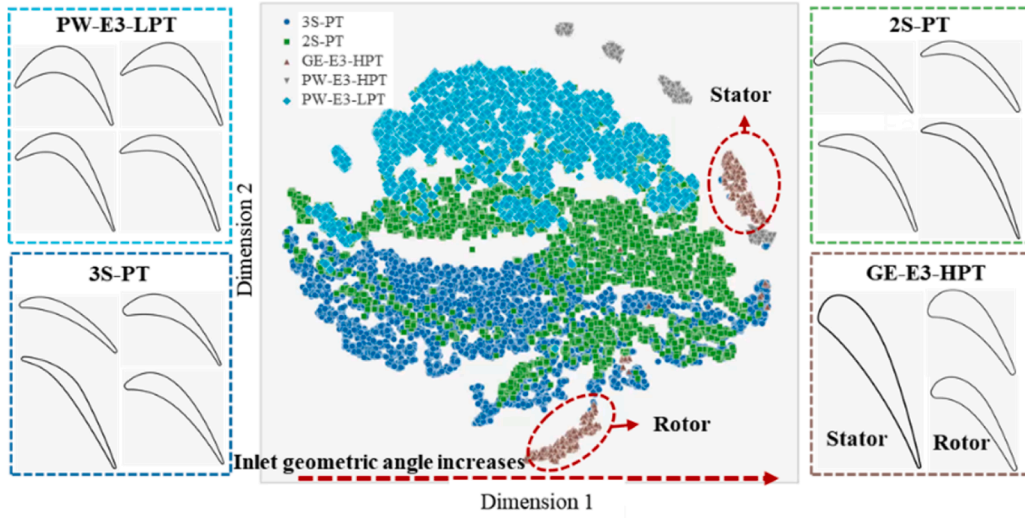


Fig. 7. t-SNE clustering distribution of various turbine blades.

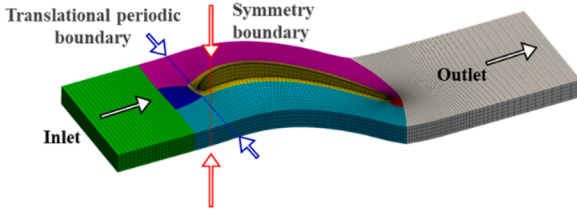


Fig. 8. Numerical calculation model for turbine blades.

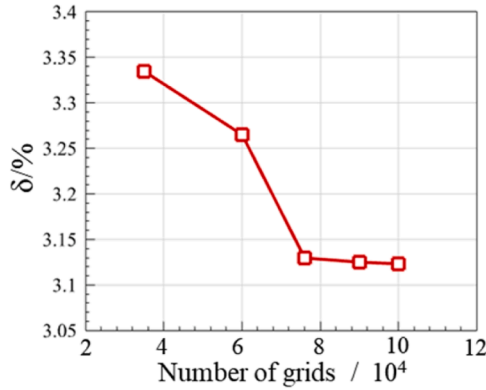


Fig. 9. Effect of grid number on the total pressure loss coefficient.

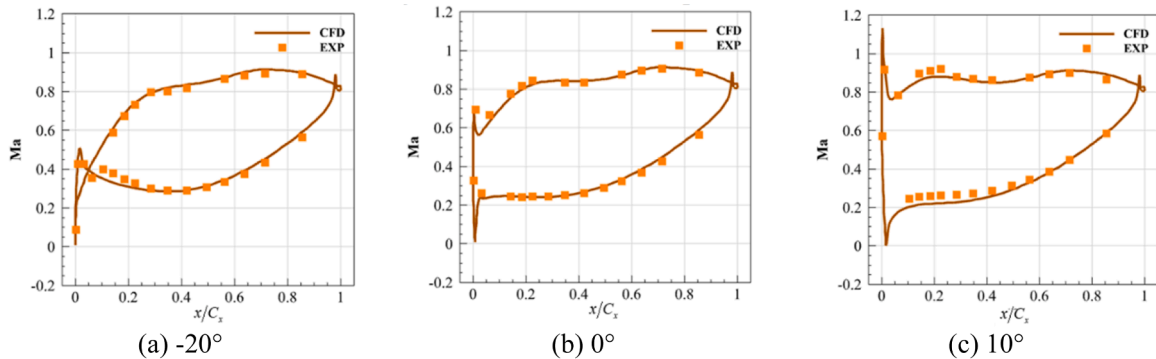


Fig. 10. Comparison between numerical and experimental results.

turbines, as well as flow regimes from subsonic to transonic.

Based on the aforementioned parameter ranges, 20,000 feature combinations were generated using the Latin Hypercube Sampling (LHS) method. To handle the massive computational burden of evaluating tens of thousands of samples, this study developed a robust, fully automated batch-calculation framework using Python scripts to directly drive the ANSYS platform in the background. Specifically, this framework sequentially invokes template-based automated meshing, runs the ANSYS CFX pre-processor and solver in batch mode, and automatically extracts the required flow field distributions and aerodynamic performance parameters during the post-processing stage. This automated approach not only tremendously accelerates the dataset construction process but also ensures strict standardization across all generated data samples. Subsequently, batch calculations were performed via the validated automated CFD workflow to construct complete, high-fidelity input-output sample pairs. Finally, the dataset was randomly partitioned into a training set of 16,000 samples and a validation set of 4000 samples, maintaining a 4:1 ratio. These sets are utilized for deep neural network parameter optimization and generalization performance assessment, respectively. The dataset features a broad distribution across geometric solidity, aerodynamic operating conditions, and Mach number ranges. This provides data support for training high-precision performance prediction models with physics-enhanced characteristics.

4. Prediction result and discussion

To evaluate the prediction performance of the proposed physics-enhanced data-driven model across a wide operating range, this

chapter conducts analysis and validation from three perspectives: training strategy, prediction accuracy, and generalization capability. First, key evaluation metrics are defined to quantify the prediction performance. Second, specific training strategies are detailed, followed by a comparative study against traditional data-driven models. Subsequently, prediction results from different neural networks on the validation set are analyzed. Finally, typical turbine blade profiles are selected as test cases. These cases focus on examining the model's capability to capture physical evolution laws under wide operating conditions, thereby confirming the effectiveness and engineering utility of the proposed method.

4.1. Evaluation metrics and training metrics

To quantitatively evaluate the prediction accuracy of the physics-enhanced model, this section establishes an evaluation metric method. The prediction tasks cover two-dimensional flow field distributions, specifically the isentropic Mach number distribution, and scalar performance metrics such as the total pressure loss coefficient. This validates the model's accuracy and stability across diverse conditions, enabled by the proposed physical constraints and non-dimensionalization.

Regarding the selection of evaluation metrics, the coefficient of determination (R^2), Mean Relative Error (MRE), and Mean Absolute Error (MAE) are adopted. These metrics quantify the deviation between model predictions and CFD calculations. The specific mathematical expressions are as follows:

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (1)$$

$$MRE = \frac{1}{N} \sum_{i=1}^N \frac{\|\hat{y}_i - y_i\|_2}{\|y_i\|_2} \quad (2)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (3)$$

In these equations, $\hat{y}_i$ and y_i represent the model predicted values and the CFD values, respectively. N denotes the total number of samples in the dataset, and $\|\cdot\|_2$ represents the L2 norm. Specifically, MRE is utilized to evaluate the prediction accuracy of the loading distribution, while MAE is used to assess the prediction accuracy of the total pressure loss coefficient.

Regarding the training strategy and hardware configuration, the model is implemented using the PyTorch. All computations are performed on an NVIDIA GeForce RTX 4090 GPU computing platform to ensure efficient training and stable convergence. The Adam algorithm is selected as the optimizer for parameter updates. The total number of training epochs is set to 800, with an initial learning rate of 0.001. Additionally, a piecewise learning rate decay mechanism is introduced. When the training process reaches 80% completion, the learning rate is reduced to 0.0001. This adjustment aims to fine-tune model parameters to precisely approximate the global optimal solution. A mini-batch stochastic gradient descent strategy is adopted during training. The batch size is set to 2048 to achieve an optimal balance between computational efficiency and the smoothness of gradient estimation. The construction of the loss function plays a decisive role in the model optimization process. For the regression tasks of flow field loading reconstruction and aerodynamic performance prediction, the Mean Squared Error (MSE) is uniformly adopted as the loss function.

4.2. Comparison of prediction accuracy of different models

To validate the efficacy of the proposed method, a comparative study is conducted between the P-ResUNet model and various baseline

models. The benchmarks encompass traditional pure data-driven models (MLP [40], UNet, ResUNet) as well as their corresponding physics-enhanced variants (P-MLP, P-UNet). As illustrated in Fig. 2, these physics-enhanced architectures consist of two cascaded sub-neural networks: a flow field prediction model and a performance prediction model.

Specifically, the input for the flow field prediction model integrates geometric and aerodynamic parameters of the two-dimensional turbine blade. It comprises two channels encoding blade geometry (represented by 256 normalized coordinates) and three channels representing aerodynamic parameters: exit isentropic Mach number, inlet incidence angle, and relative pitch. The output is a single-channel distribution of the blade surface isentropic Mach number. Subsequently, the performance prediction model takes this Mach number distribution as input to map and output the single-channel total pressure loss coefficient. To ensure fairness and rigor in the comparison, all models are evaluated using identically constructed training and validation sets. Furthermore, they strictly adhere to the iterative optimization and performance evaluation strategies established in Section IV A.

Table 1 summarizes the quantitative data for loading distribution reconstruction and total pressure loss coefficient prediction across all models. Data analysis yields several key conclusions:

- (1) The physics-enhancement mechanism significantly improves prediction accuracy. Physics-enhanced models generally outperform their traditional data-driven counterparts in predicting scalar performance parameters. Taking the MLP architecture as an example, the introduction of physics enhancement (P-MLP) drastically reduces the MAE on the validation set from 2.84% to 0.35%. This indicates that incorporating physical information provides explicit prior constraints for the neural network, thereby greatly reducing the optimization difficulty when fitting complex physical mappings.
- (2) The synergy between network architecture and physical information is significant. Among all baseline models, the ResUNet-based architecture achieves the best performance in capturing subtle flow features and predicting performance metrics. This fully validates the architectural advantages of combining residual structures with the U-Net framework for nonlinear flow feature extraction and spatial resolution preservation.
- (3) Physical constraints significantly enhance the model's generalization capability and robustness. The accuracy of physics-enhanced models on the validation set is highly consistent with that on the training set. This contrasts sharply with the evident overfitting observed in traditional models on the validation set. It provides strong evidence that introducing physical priors effectively mitigates overfitting, thereby significantly enhancing prediction stability under unseen operating conditions.

To further analyze the statistical characteristics of prediction errors, Fig. 11 presents the Kernel Density Estimation (KDE) of prediction errors for the three physics-enhanced models. Regarding the loading distribution prediction shown in Fig. 11(a), the error distribution curve of P-ResUNet exhibits the narrowest distribution and highest peak across

Table 1

Comparison of prediction results among different models.

Model	Training set		Validation set	
	MRE of Ma (%)	MAE of δ (%)	MRE of Ma (%)	MAE of δ (%)
MLP		0.76		2.84
P-MLP	1.54	0.16	2.16	0.35
UNet	–	1.13	–	1.87
P-UNet	1.60	0.26	1.73	0.31
ResUNet	–	0.60	–	1.52
P-ResUNet	0.94	0.17	1.16	0.26

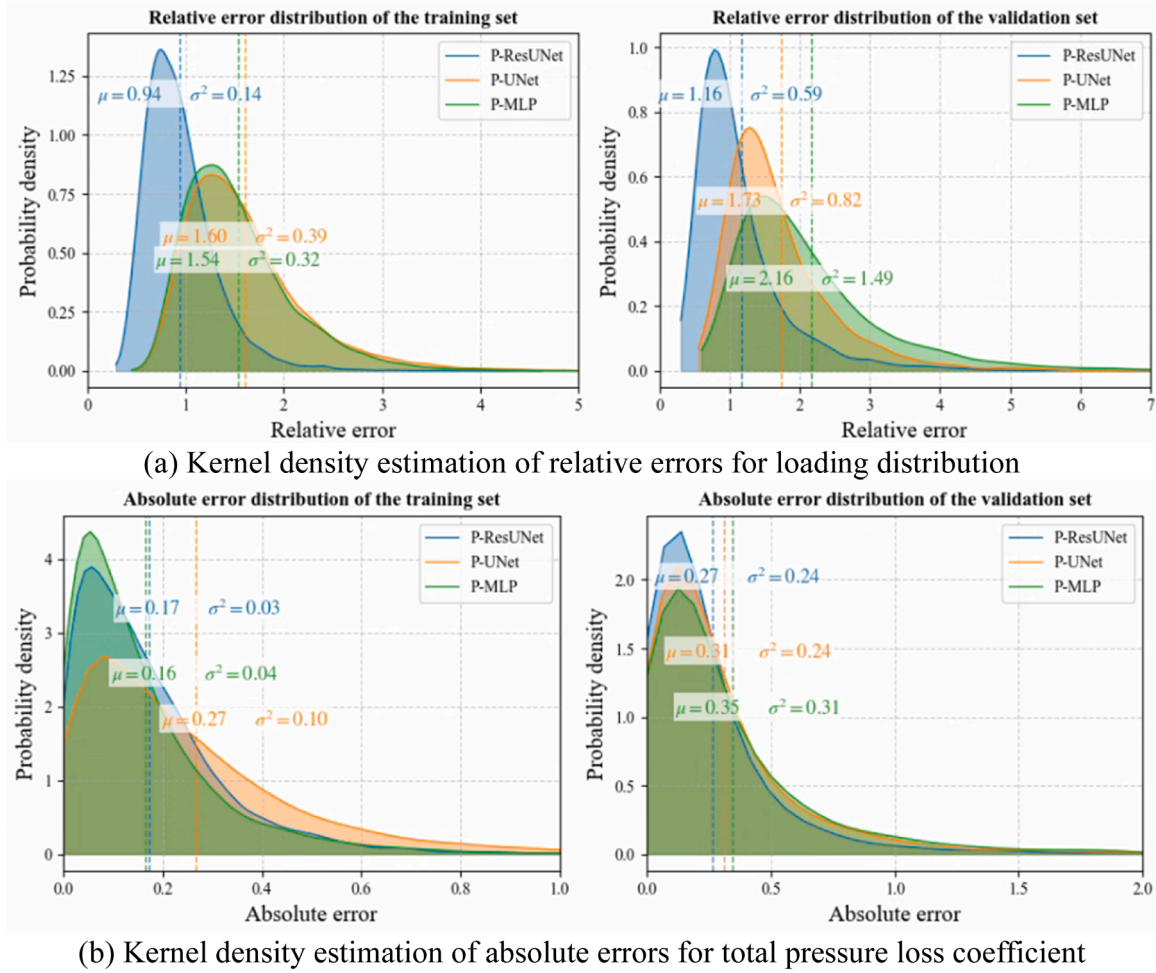


Fig. 11. Statistical distribution comparison of prediction errors among various models.

both training and testing sets. This characteristic reflects its superior prediction accuracy. Significant differences are observed in the prediction error distribution for the total pressure loss coefficient, as shown in Fig. 11(b). Although P-MLP demonstrates extremely high fitting sharpness on the training set, its distribution significantly broadens on the test set, revealing a tendency toward overfitting. In contrast, P-ResUNet maintains the narrowest and most concentrated distribution morphology on the test set. Its prediction errors cluster tightly around

zero, indicating exceptional stability when handling unseen data.

In summary, P-ResUNet achieves the optimal balance between high-precision fitting and strong generalization capability. This success is attributed to the construction of the physics-enhanced prediction method, fully verifying its critical role in enhancing engineering value and supporting the intelligent design framework.

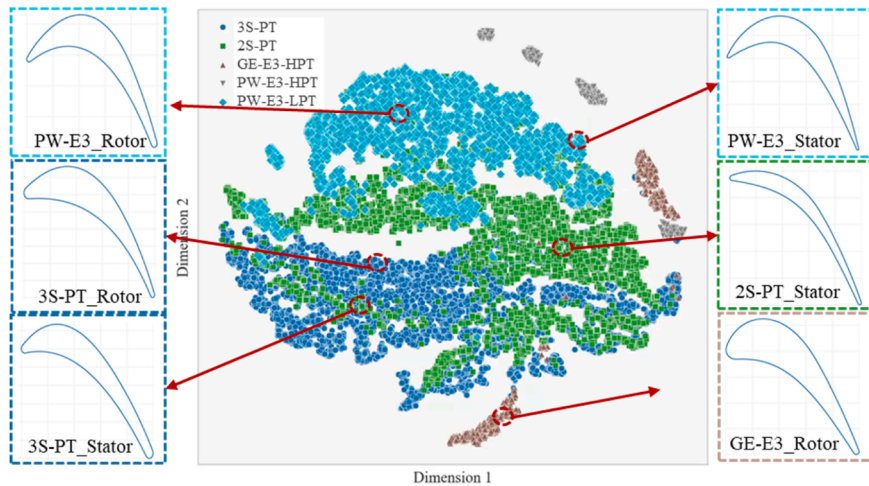


Fig. 12. Visualization of test samples in the turbine database.

4.3. Prediction results for validation set samples

Following the statistical accuracy comparison of different models presented earlier, this section focuses on the specific performance of the P-ResUNet model on representative samples from the validation set. Flow field visualizations are employed to evaluate the model's capability to capture complex flow features.

Fig. 12 illustrates the distribution of the selected test blade profiles within the t-SNE feature space of the turbine database. It is observed that the test samples exhibit high dispersion and broad coverage within the feature space. They not only span multiple independent geometric clusters but also encompass various typical geometric types, ranging from high-solidity vanes to low- and medium-solidity blades. Combined with significant variations in incidence angle and exit Mach number, these test cases ensure the diversity of the validation process across both geometric and aerodynamic dimensions. Consequently, the scope of application of the model can be effectively verified.

Fig. 13 presents a comparison between the predicted isentropic Mach number distributions and total pressure loss coefficients against the CFD values. Corresponding prediction deviations are also quantified. In the figures, red solid lines represent the CFD calculations, while blue dashed lines denote the model predictions.

The comparison indicates that, the P-ResUNet model demonstrates superior accuracy and robust generalization performance when handling wide-range loading distribution prediction tasks. Under the vast majority of operating conditions, the prediction curves maintain high agreement with the CFD baseline curves across the entire chord length. Notably, in regions where the flow remains relatively attached, such as the mid-span and pressure surface, prediction errors generally converge to within 1.5%. Regarding aerodynamic performance metrics, excluding a few extreme off-design cases involving severe flow separation, the absolute error of the total pressure loss coefficient for most samples is controlled below 0.2%. This observation remains highly consistent with the statistical evaluation conclusions presented in the previous section.

An analysis of loading distributions under different conditions reveals the model's capability to capture complex physical flow mechanisms. First, under subsonic flow conditions, as shown in Fig. 13(a) and

Fig. 13(e), the model accurately reproduces the fundamental acceleration and deceleration characteristics of the flow field. Second, under transonic conditions, the model effectively identifies and captures sudden loading changes and strong adverse pressure gradient structures induced by passage shocks, as shown in Fig. 13(b) and Fig. 13(d). Crucially, for complex nonlinear flow fields accompanied by local boundary layer separation, as shown in Fig. 13(f), the model accurately reconstructs the specific morphology of the loading distribution. This provides strong evidence of the physics-enhanced architecture's exceptional adaptability to complex flows.

It is worth emphasizing that, although the selected test profiles are distributed in disjoint, discrete regions within the feature space, the model maintains highly stable prediction accuracy. This suggests the model has effectively learned the underlying nonlinear flow correlations rather than simply memorizing the training data. In summary, the superior performance on representative validation samples further confirms the model's robustness against variations in geometric and aerodynamic conditions. This predictive capability, combining high fidelity with efficiency, lays a technical foundation for the subsequent intelligent optimization design.

4.4. Testing on typical turbine blade cases

To further validate the prediction accuracy of the proposed physics-enhanced model when facing unknown geometric features and wide-ranging operating conditions, two typical turbine blade profiles were selected as test cases. These blade profiles are completely independent of the training and validation sets. Their geometries are shown in Fig. 14. Blade A represents the mid-span section of a power turbine stator. Blade B corresponds to the root section of a low-pressure turbine rotor, characterized by a large turning angle. Based on these distinct profiles, this section evaluates the prediction capability regarding variations in relative pitch, exit isentropic Mach number, and inlet incidence angle.

Fig. 15 compares the predicted surface loading distributions and total pressure loss coefficients with CFD results under fixed exit isentropic Mach number and inlet incidence. The analysis shows that the model accurately captures the evolution of loading distribution induced by pitch variations. For Blade A, as relative pitch increases, the peak

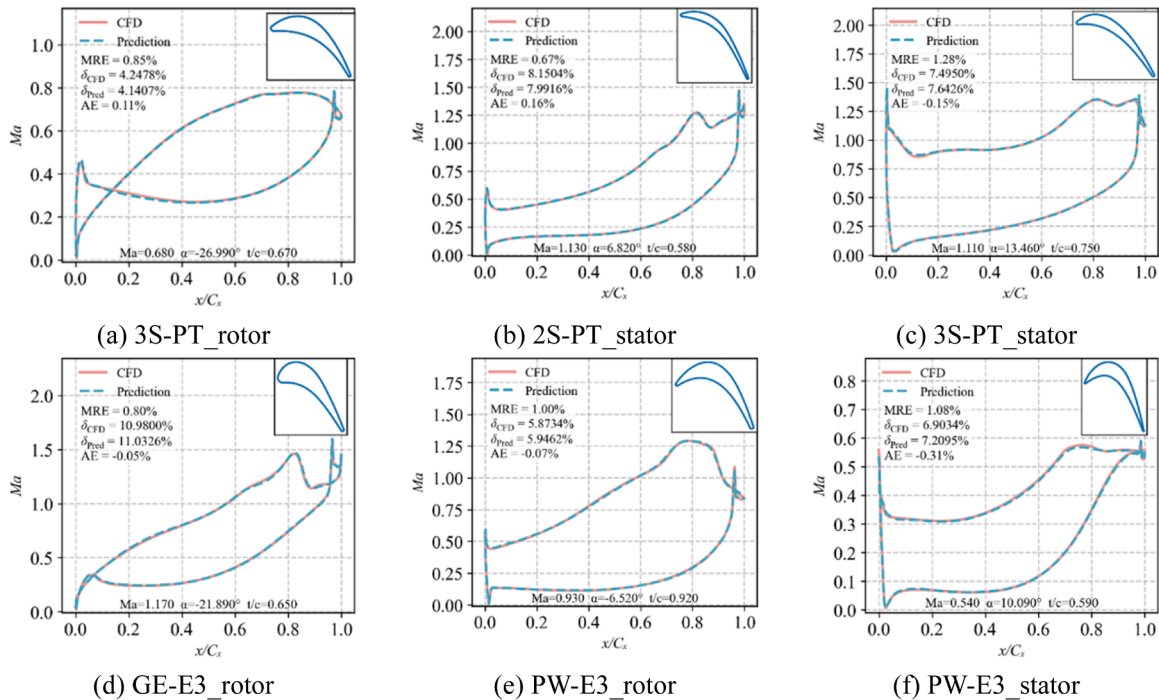


Fig. 13. Comparison of prediction results for different turbine blades in the validation set.

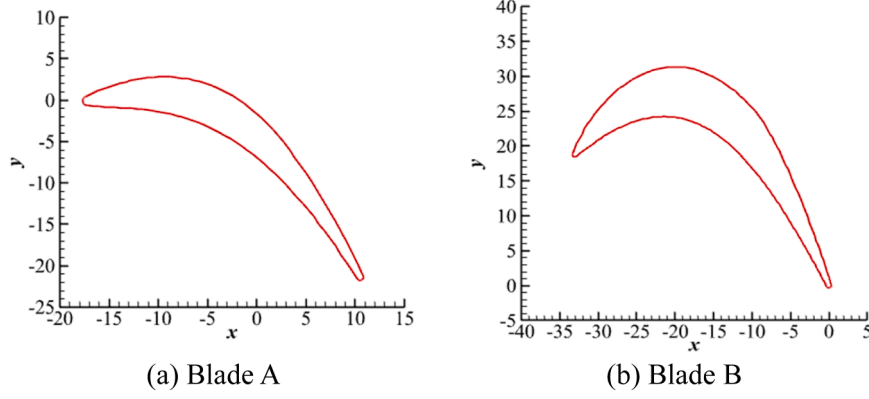


Fig. 14. Schematic of typical turbine blade profiles.

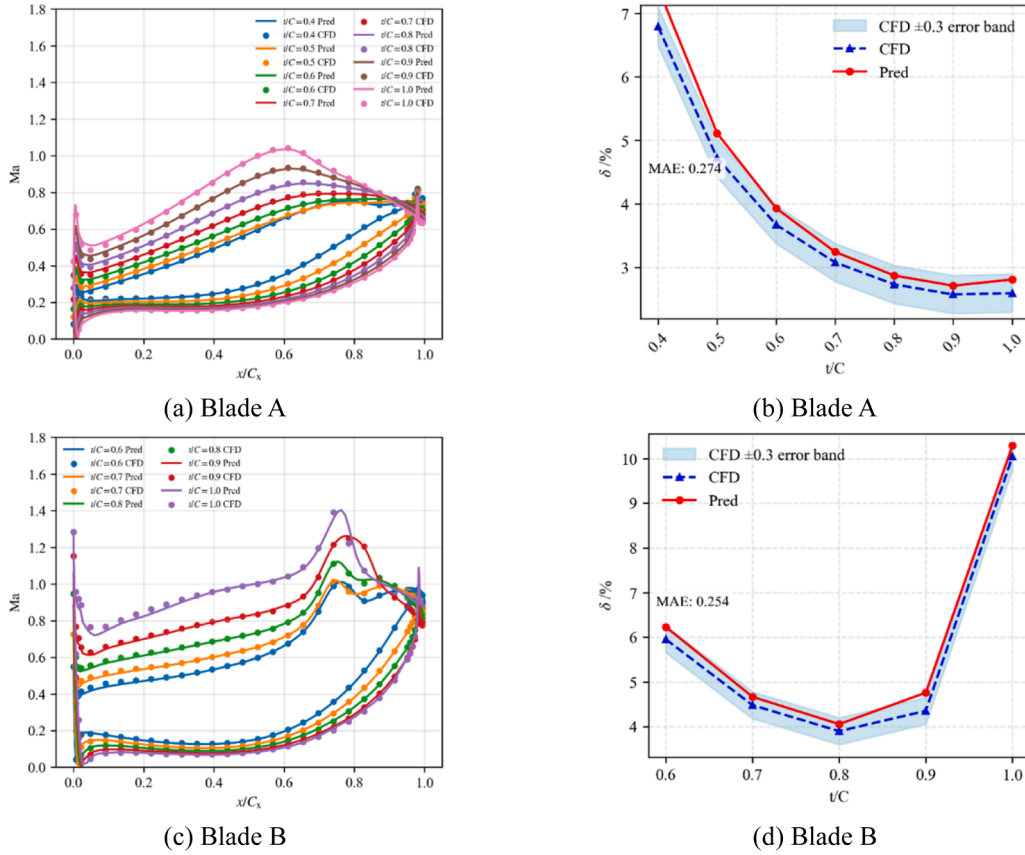


Fig. 15. Variations of performance parameters with relative pitch for a specific turbine blade.

loading rises significantly and shifts upstream. The model accurately predicts the monotonic decrease in total pressure loss coefficient, achieving MAE of only 0.254%. Regarding Blade B, the model captures the strengthening of the suction-side peak loading. It also successfully identifies the complex physical process where shock intensity first weakens and then strengthens as pitch increases. Notably, the model accurately predicts the non-linear inflection point for Blade B's total pressure loss coefficient, which first decreases and then increases significantly. It successfully identifies the optimal aerodynamic operating point at $t/C = 0.8$. This indicates that through geometric non-dimensionalization, the model has effectively grasped the intrinsic physical mechanisms of geometric similarity affecting aerodynamic performance.

Fig. 16 illustrates the prediction capability for loading distribution

and loss characteristics across a wide range from subsonic to transonic regimes, with fixed relative pitch and inlet incidence. In the subsonic region, the model's predictions almost perfectly overlap with CFD results. When the flow enters the transonic regime ($Ma_1^{is} > 0.9$), the model successfully captures the onset, formation, and downstream movement of the throat shock. It maintains excellent accuracy even in high-gradient loading regions around the shock. Regarding performance metrics, the total pressure loss coefficient exhibits significant non-linear abrupt changes with exit Mach number. When $Ma_1^{is} \leq 0.9$, the variation in loss coefficient is relatively flat. However, when $Ma_1^{is} > 0.9$, the loss coefficient rises sharply due to shock waves. The prediction model clearly defines the physical boundary between subsonic and transonic flows, further verifying its adaptability to aerodynamic similarity criteria.

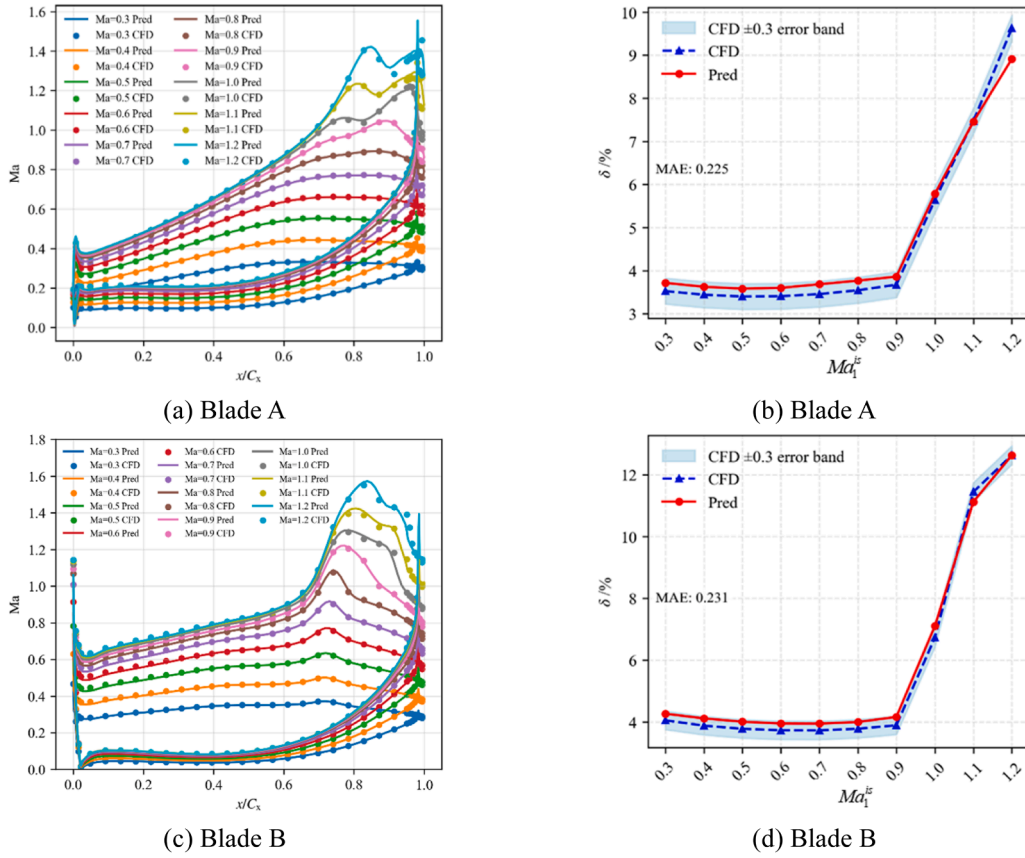


Fig. 16. Variations of performance parameters with exit isentropic Mach number for a specific turbine blade.

Fig. 17 displays prediction performance under varying inlet incidence angles ranging from -30° to 10° , with fixed relative pitch and exit Mach number. The model provides accurate feedback on the variation of suction-side peaks and the evolution of pressure-side loading under different incidence angles. Particularly under positive incidence conditions, the model precisely reconstructs the onset location and morphology of potential flow separation on the suction side of Blade B. Regarding performance trend prediction, Blade A exhibits broad operational adaptability, maintaining low total pressure loss coefficients across all incidence angles. Conversely, Blade B suffers a sharp increase in loss under positive incidence due to significant flow separation. This flow deterioration is acutely captured by the model. These analyses confirm the model's strong stability in handling complex inlet conditions and non-linear mappings of separated flows.

In summary, tests were conducted on two unseen typical profiles under continuous variations of relative pitch, exit isentropic Mach number, and inlet incidence angle. The results indicate that the proposed physics-enhanced model not only maintains high accuracy within the parameter space covered by training data but also accurately predicts aerodynamic characteristics of unseen profiles under wide operating conditions. The demonstrated operational adaptability of the model lays a solid technical foundation for the intelligent design of turbine blades under complex, real-world conditions.

5. Cases validation of intelligent design methods

To verify the feasibility and efficiency of the proposed intelligent design framework in practical engineering applications, this section conducts typical verification cases under both single and multiple conditions. As shown in Fig. 5(a), the framework constructs a design path from initial requirements to the final optimal blade profile. The verification process follows a progressive strategy. First, a correlation analysis

of design variables is performed to provide physical insights and define the optimization boundaries. Next, single-condition forward and inverse designs are executed to demonstrate geometric reconstruction and performance enhancement at specific design points. The scope is then expanded to multi-condition scenarios to verify loading matching accuracy and robustness across broad operating ranges. Finally, a quantitative analysis of computational efficiency is provided to highlight the resource-saving advantages.

5.1. Correlation analysis of design variables based on the analysis of variance

Before conducting the intelligent optimization design, a correlation analysis of the design variables based on the analysis of variance [41] was conducted. This step helps analyze how turbine blade design variables affect aerodynamic performance and provides a reasonable design space for subsequent optimization. Following the blade parameterization method shown in Fig. 6, we built a dataset based on Blade A. During this process, the cascade pitch and inlet flow angle remained fixed. We sampled the axial chord, correlation coefficients, other angular design variables, and control point coefficients within $\pm 5\%$ of their baseline values. The sampling ranges for the leading edge radius and trailing edge radius were set to $[0.3 \text{ mm}, 2.0 \text{ mm}]$ and $[0.3 \text{ mm}, 0.6 \text{ mm}]$, respectively.

Fig. 18 shows the variance contribution charts and main effect curves obtained from the analysis of variance. To clearly reveal the impact weight of each variable, the analysis process was divided into two steps.

First, in the analysis including all variables (as shown in Fig. 18(a) and (b)), the trailing edge radius R_{te} emerges as the most globally significant design variable. This occurs even though its sampling variation range is relatively small. The main effect analysis indicates that a larger trailing edge radius leads to a sharp upward trend in the total pressure loss coefficient. This reflects the essential principle that the trailing edge

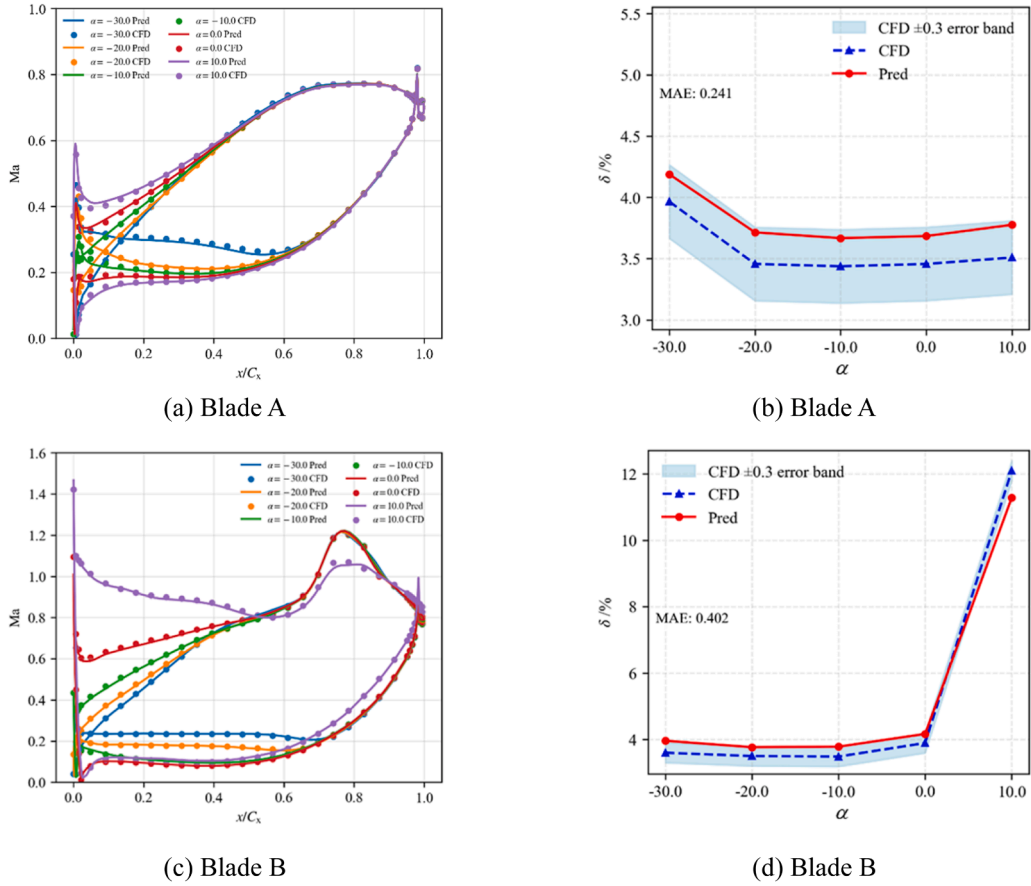


Fig. 17. Variations of performance parameters with angle of attack for a specific turbine blade.

thickness directly determines the magnitude of the wake mixing loss. Therefore, in the subsequent optimization design, the trailing edge radius is fixed at 0.3 mm by considering the minimum constraints of actual manufacturing processes. It is no longer treated as an optimization variable.

Second, the extremely high sensitivity of the trailing edge radius masks the effects of other geometric parameters to some extent. Thus, further analysis was conducted after fixing the trailing edge radius (as shown in Fig. 18(c) and (d)). The results show that among the remaining design variables, the effective exit angle β_2 and the axial chord C_x have the most prominent impact weights. Specifically, the effective exit angle directly determines the exit mass flow rate of the cascade. To achieve strict flow matching, its variation range must be constrained within $\pm 0.5^\circ$ in the subsequent optimization. Meanwhile, the axial chord significantly affects the flow acceleration and load distribution within the blade passage by altering the relative pitch. The next most sensitive variables are the control points of the second Bézier curve on the suction surface. This demonstrates that the profile curvature of the mid-to-rear suction surface is crucial for controlling boundary layer development and flow separation. The remaining geometric design variables have a relatively small impact on the total pressure loss.

5.2. Testing of inverse design case under single-condition

5.2.1. Verification of inverse design case

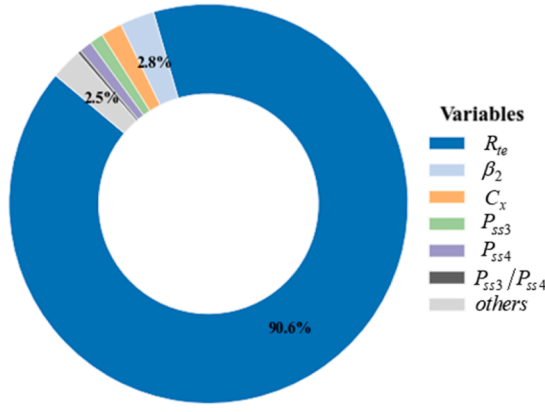
A benchmark test for a typical design point under a single condition is conducted. The case specifies fixed conditions: the cascade relative pitch is fixed at 23.32 mm, the inlet total pressure and total temperature are set to 350 kPa and 1000 K, respectively, and the outlet static pressure is 252.32 kPa. These correspond to an exit isentropic Mach number of

0.70 and an inlet incidence angle of 0° . Under these boundary conditions, key geometric parameters, including axial chord length, leading-edge radius, tangential chord angle, and various profile control points, are selected as design variables. Optimization is performed within the design space defined in Table 2. The objective function is set to minimize the deviation between the predicted loading of the candidate profile and the target loading distribution shown in Fig. 18(b) (indicated by orange scatter points).

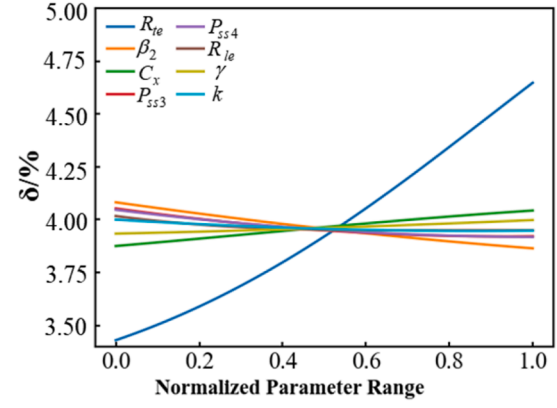
The optimization algorithm converges after 100 iterations. Fig. 19 presents the final generated inverse design profile and its loading distribution evaluation. As seen in Fig. 19(a), the generated profile features a smooth, continuous camber line and reasonable thickness distribution, adhering to aerodynamic design principles. Fig. 19(b) further compares the predicted loading distribution of the inverse design with the target distribution. The two show extremely high agreement across the entire chord length, with the model accurately capturing both the magnitude and axial position of the suction-side peak loading. To further validate reliability, a CFD simulation was performed on the inversely designed profile. Results show that the actual loading distribution obtained from CFD essentially overlaps with the model prediction. Regarding aerodynamic performance, the CFD-calculated total pressure loss coefficient is 3.45%, while the model prediction is 3.47%, maintaining a very low error margin. These results strongly prove that the proposed framework can efficiently reconstruct turbine blade geometry based on given aerodynamic loading requirements.

5.2.2. Verification of forward design case

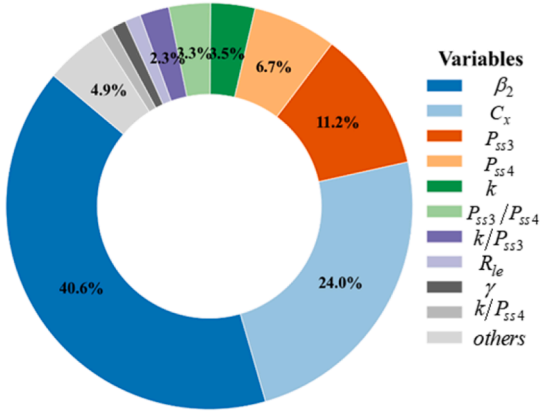
To verify the intelligent design framework's capability to enhance aerodynamic performance, a single-condition forward design test is conducted. Unlike the inverse design task, this case directly targets the minimization of the total pressure loss coefficient. The test is based on



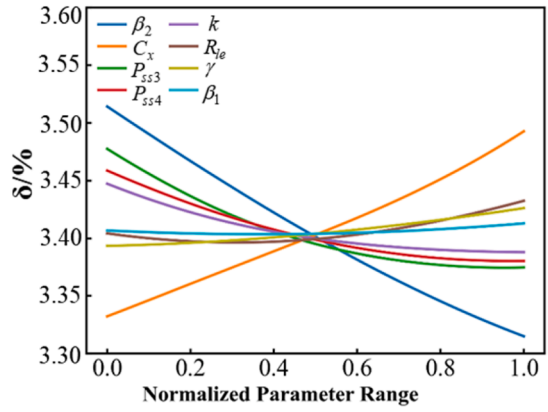
(a) Variance contribution chart considering the trailing edge radius



(b) Main effect curves considering the trailing edge radius



(c) Variance contribution chart without considering the trailing edge radius



(d) Main effect curves without considering the trailing edge radius

Fig. 18. Correlation analysis of design variables based on the analysis of variance.

Table 2

Ranges of design variables for single-condition inverse design.

Parameter	Range	Parameter	Range
$C_x/(mm)$	[20,50]	$\theta/(^\circ)$	[5,10]
$R_{te}/(mm)$	[0.3,2.0]	k	[0.2,0.4]
$\gamma/(^\circ)$	[20,60]	$e_2/(^\circ)$	[5,10]
$\beta_1/(^\circ)$	[30,90]	P_{ss}	[0.1,0.9]
$\varepsilon_1/(^\circ)$	[5,20]	P_{ps}	[0.1,0.9]
$\beta_2/(^\circ)$	[20,30]		

specific aerodynamic boundary conditions: pitch, inlet total pressure, total temperature, and outlet static pressure are identical to the inverse design case, corresponding to an exit isentropic Mach number of 0.70, with the inlet flow angle set to 9° . Under these conditions, the optimization is performed on key geometric variables listed in Table 3, such as leading-edge radius and inlet geometric angle, to explore the optimal aerodynamic profile. In this process, specific geometric constraints were enforced: the trailing edge radius was kept constant, and the effective exit angle was restricted to vary within a narrow range to ensure precise control of the mass flow rate.

Using the inversely designed profile from Fig. 19(a) as the baseline, the optimization algorithm drives the intelligent prediction model for performance iteration. After 100 generations, the algorithm converges to a global optimum. Fig. 20(a) contrasts the blade profiles before and after optimization. It is observed that the optimized profile exhibits significantly reduced overall thickness, presenting a slender geometric characteristic. This typically helps delay boundary layer transition on

the suction side and suppress profile friction losses.

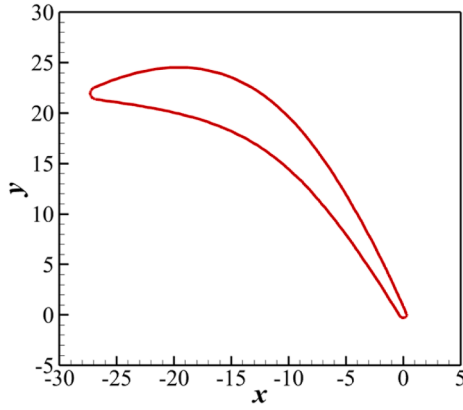
Table 4 details the quantitative comparison of total pressure loss coefficients. It indicates that the coefficient decreases from 3.47% for the baseline design to 3.18% for the optimized profile, representing a relative aerodynamic performance improvement of 8.36%. Fig. 20(b) further reveals details of the loading distribution improvement. Compared to the baseline, the optimized suction-side loading distribution is smoother, with the peak position moderately shifted downstream. This effectively suppresses excessive local expansion during flow acceleration, thereby significantly reducing flow losses within the passage.

5.3. Testing of inverse design vase under multiple-conditions

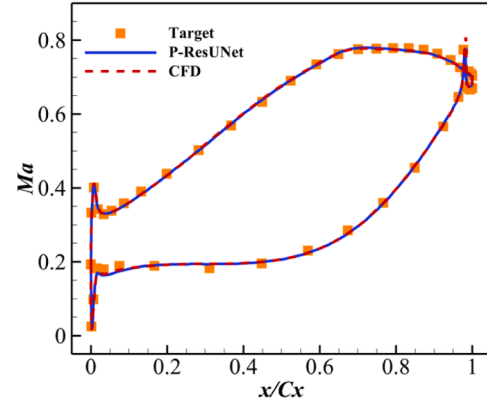
5.3.1. Verification of inverse design case

To further verify the robustness of the constructed intelligent design framework under a wide range of operating conditions, multi-point inverse design tests were conducted. Unlike the single-condition test, the objective here is to examine whether a single blade profile can satisfy pre-set target loading distributions goals under multiple inflow conditions.

First, the multi-point inverse design case aims to verify whether a single profile can meet target loading requirements across three distinct inlet incidence angles: -20° , 0° , and 10° . The fixed operating parameters are set as follows: cascade relative pitch is 18.78 mm; inlet total pressure and total temperature are 400 kPa and 1500 K, respectively; outlet static pressure is 249.4 kPa, corresponding to an exit isentropic Mach number of 0.85. The adjustment ranges for geometric design variables remain



(a) Inversely designed turbine



(a) Validation of inverse design results

Fig. 19. Inverse design results under a single operating condition.

Table 3

Ranges of design variables for single-condition forward design.

Parameter	Range	Parameter	Range
$R_{le}/(\text{mm})$	[0.3,1.0]	$\theta/(^{\circ})$	[5,10]
$\beta_1/(^{\circ})$	[74,88]	k	[0.2,0.3]
$\epsilon_1/(^{\circ})$	[10,24]	P_{ss}	± 0.1
$\beta_2/(^{\circ})$	[24.5,25.5]	P_{ps}	± 0.1

consistent with Table 2 in Section A. The optimization algorithm converges after 100 iterations.

Fig. 21 presents the final generated inverse design profile and the evaluation of its loading distribution. As shown in Fig. 21(a), the inversely designed profile exhibits geometric features adaptable to wide incidence variations, such as a moderate leading-edge radius and optimized camber line curvature. These features provide a basis for suppressing separation losses under different incidence angles.

Regarding the results in Fig. 21, the framework demonstrates excellent loading matching capability across different incidence angles. Under the large negative incidence condition of -20° , the model accurately captures the characteristic where suction-side loading at the leading edge is lower than that on the pressure side. The suction-side loading morphology aligns highly with the target, demonstrating good adaptability to extreme inlet conditions. At the design incidence of 0° , the predicted curve almost perfectly overlaps with the target distribution, with errors in peak magnitude and axial position reduced to minimal levels. Even under the large positive incidence of 10° , the model

still reproduces the target loading trend with high precision, particularly in the mid-to-aft region of the suction side. CFD validation confirms that the actual loading distribution essentially coincides with the predicted values. This fully proves that the intelligent design framework is capable not only of precise matching at a single design point but also of achieving wide-adaptability design by adjusting key geometric parameters.

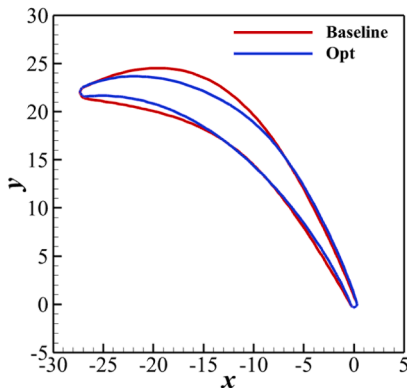
6. Verification of forward design case

Subsequently, a multi-point forward design verification was conducted to demonstrate the framework's capability to enhance aerodynamic performance. The objective is to minimize the total pressure loss coefficient under three different inlet flow angles: 10° , 20° , and 40° . The test is based on specific aerodynamic boundary conditions where pitch, inlet total pressure, total temperature, and outlet static pressure are identical to the inverse design case, corresponding to an exit isentropic Mach number of 0.85. Under these conditions, the optimization is performed on key geometric variables listed in Table 5. In this process, specific geometric constraints were enforced: the trailing edge radius

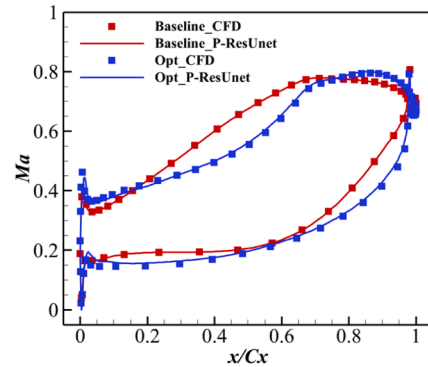
Table 4

Comparison of forward design optimization results for the total pressure loss coefficient under single - condition.

	P-ResUNet (%)	CFD (%)	AE (%)
Baseline design	3.45	3.47	0.02
Optimized design	3.14	3.18	-0.04
Relative improvement	8.99	8.36	



(a) Comparison of forward-designed blades



(b) Validation of optimization results

Fig. 20. Forward design results.

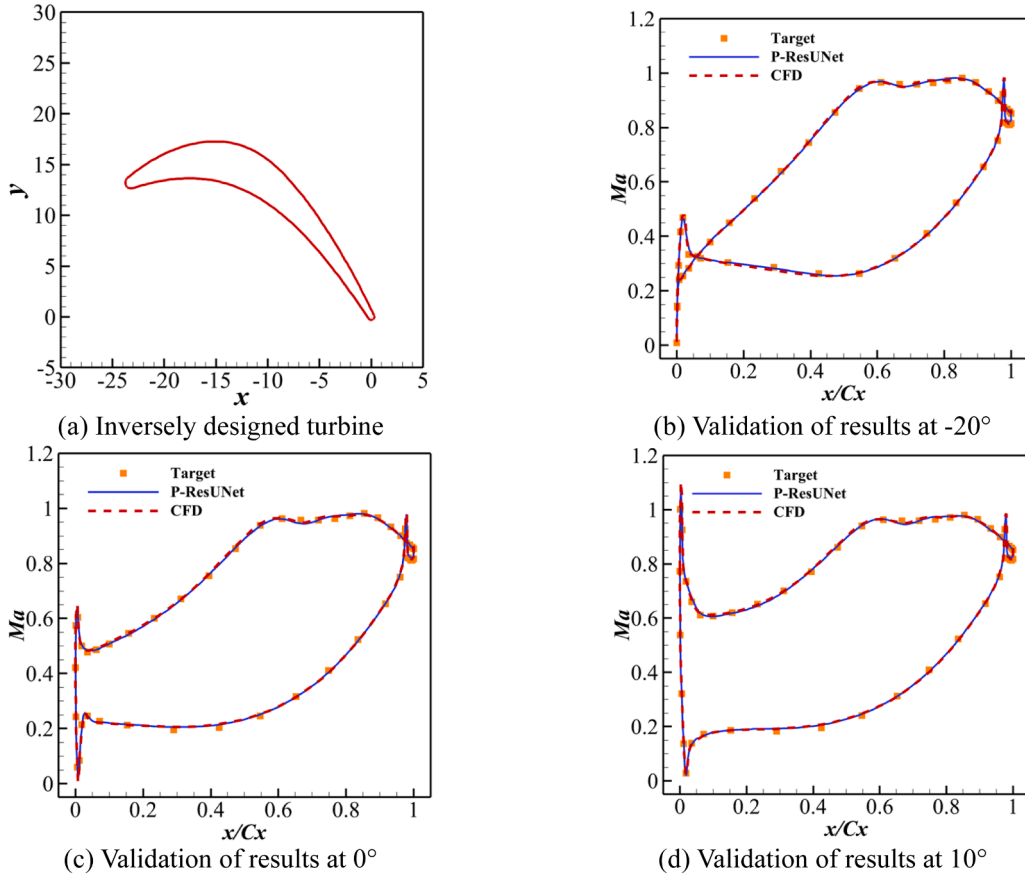


Fig. 21. Inverse design results under multi-point operating conditions.

was kept constant, and the effective exit angle was restricted to vary within a narrow range to ensure precise control of the mass flow rate.

Table 6 presents the comparison results with the baseline design, showing that the optimized profile achieves a significant reduction in the total pressure loss coefficient across all tested conditions. Specifically, CFD-calculated losses decreased from 3.82% to 3.60% at 10° , from 3.60% to 3.50% at 20° , and from 3.69% to 3.48% at 40° . Meanwhile, prediction errors remained consistently low, with the MAE controlled within 0.35%.

Geometric and flow field details of the optimization results are shown in Fig. 22. The profile comparison in Fig. 22(a) reveals that, compared to the baseline, the optimized profile features a significantly increased leading-edge radius and slightly increased blade thickness. These geometric changes typically improve flow around the leading edge under off-design conditions, thereby enhancing adaptability to different inlet flow angles. Further observation of the surface isentropic Mach number distributions in Fig. 22(b)-(d) confirms that the physics-enhanced model's prediction curves maintain high agreement with CFD values across all conditions. The model not only accurately captures loading characteristics on both suction and pressure sides but also verifies that the optimized profile maintains ideal loading morphology across a broad range of inflow angles. Compared to the baseline design,

the optimized suction-side loading distribution is smoother, with the peak position moderately shifted downstream and the suction-side curve appearing fuller. This effectively suppresses excessive local expansion during acceleration, significantly reducing flow losses within the passage. These results not only validate the accuracy of the physics-enhanced model in guiding the optimization process but also fully demonstrate that the method can achieve significant and stable aerodynamic performance improvements in complex engineering scenarios.

6.1. Efficiency and resource consumption analysis

Following the validation of the optimization design cases under single and multiple operating conditions, this section further quantitatively analyzes the performance of the proposed intelligent design framework in conserving design resources, such as computational facilities and time.

During the turbine blade design optimization process, traditional CFD evaluations are time-consuming. Taking the cases in this paper as an example, a single CFD solution for one sample usually takes about 100 s on a 16-core server. In contrast, a single performance evaluation using the proposed P-ResUNet prediction model takes only milliseconds. Specifically, in the optimization algorithm workflow of this paper, the initial population size for single-condition forward/inverse design is 64. The population size for each iteration is 16. With 100 optimization steps, a total of 1664 performance evaluations are required. For multi-condition forward/inverse design, the initial population size is set to 128. The iteration population size remains 16. After 100 optimization steps, a total of 1728 candidate blade profiles are generated. Moreover, the multi-condition task requires evaluating 3 different inlet incidence angles or Mach number states simultaneously. Therefore, the actual total number of evaluations reaches up to 5184.

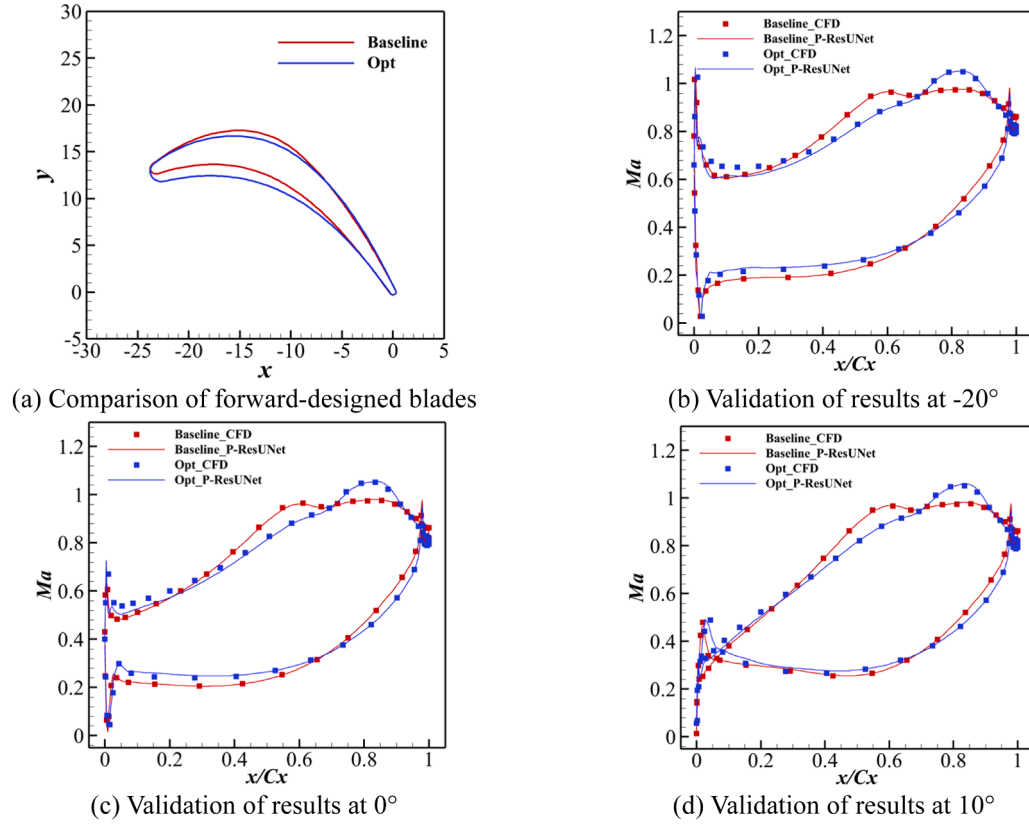
Table 5
Ranges of design variables for multi-point forward design.

Parameter	Range	Parameter	Range
$R_{le}/(\text{mm})$	[0.3,2.0]	$\theta/(\circ)$	[5,10]
$\beta_1/(\circ)$	[55,65]	k	[0.2,0.3]
$\epsilon_1/(\circ)$	[8.5,14.5]	P_{ss}	± 0.1
$\beta_2/(\circ)$	[28.5,29.5]	P_{ps}	± 0.1

Table 6

Comparison of forward design optimization results for the total pressure loss coefficient under multiple - condition.

	10°			20°			40°		
	P-ResUNet (%)	CFD (%)	AE (%)	P-ResUNet (%)	CFD (%)	AE (%)	P-ResUNet (%)	CFD (%)	AE (%)
Baseline design	3.74	3.82	−0.08	3.64	3.60	0.04	3.78	3.69	0.19
Optimized design	3.48	3.60	−0.12	3.60	3.50	0.10	3.43	3.48	−0.05
Improvement	6.95	5.76		1.99	2.78		9.26	5.69	

**Fig. 22.** Forward design results under multi-point operating conditions.

As illustrated by the data comparison in Table 7, if the traditional CFD-based design method is employed, the cumulative computational time required to complete the full optimization search would be approximately 6 h and 18 h for the single and multiple operating conditions, respectively, even on a high-performance server equipped with a 128-core CPU (AMD EPYC 7T83 64-Core Processor). Conversely, using only a standard 12-core desktop CPU (Intel Core i5-10400F), the intelligent design framework completes the single-condition forward design in just about 2 min and the inverse design in approximately 5 min. This tremendous leap in efficiency not only compresses the total design time from days to mere minutes but also drastically reduces the

reliance on expensive high-performance computing clusters during the preliminary design phase, effectively bypassing the substantial costs associated with trial-and-error physical wind tunnel experiments.

7. Conclusion

To address the challenges of time-consuming CFD evaluations and the limited generalization capability of conventional data-driven models in turbine blade design, this study proposes a physics-enhanced performance prediction method and intelligent design framework for turbine blades. By integrating prior physical knowledge with deep learning architectures, the following key conclusions are drawn:

- (1) Construction of a physics-enhanced prediction model with high precision. Distinguishing itself via physics-guided feature engineering and a knowledge-embedded architecture, the proposed method incorporates fluid dynamic similarity principles to non-dimensionalize inputs, establishing a unified representation across varying scales and operating conditions. Furthermore, the P-ResUNet architecture is employed to predict flow field distributions as intermediate physical constraints, reinforcing the learning of complex non-linear mappings. Validation results demonstrate significant superiority over traditional data-driven models, achieving a Mean Relative Error (MRE) of 1.16% for

Table 7

Comparison of time and resource consumption for intelligent design optimization.

Evaluation method and hardware configuration	Single-condition		Multi-condition	
	Forward design	Forward design	Forward design	Forward design
CFD (Server: 128-core AMD EPYC 7T83)	~ 6 h		~ 18 h	
P-ResUNet (Desktop: 12-core Intel i5-10400F)	~ 2 min	~ 5 min	~ 5 min	~ 10 min

flow field reconstruction and a Mean Absolute Error (MAE) of 0.26% for total pressure loss coefficients.

- (2) Verification of robust generalization and physical consistency. The model demonstrates excellent engineering applicability, particularly in capturing flow characteristics under varying geometries and operating conditions. Even with unseen blade profiles, it accurately predicts the evolution of aerodynamic performance with respect to changes in pitch, Mach number, and incidence angle. Notably, the model exhibits high physical consistency in capturing critical flow phenomena, such as transonic shock discontinuities and flow separation at high incidence angles. This confirms that the model has learned the underlying universal flow mapping relationships rather than merely memorizing data distributions.
- (3) Establishment of a unified intelligent design framework. By utilizing the high-precision predictor as a proxy evaluator, an efficient framework integrating loading-driven inverse design and performance-driven forward optimization is established. This framework reduces single-point evaluation time to the millisecond level while maintaining CFD-comparable accuracy. Application results confirm that the framework can precisely reconstruct geometries matching target loadings and significantly reduce total pressure loss under strict constraints. This provides a powerful tool for the rapid iteration and automated design of complex turbine blades.

In summary, through the fusion of similarity theory and deep learning, this study effectively overcomes the applicability bottlenecks of traditional models regarding wide geometric scales and broad operating conditions. It offers a feasible pathway for realizing the wide-range, intelligent, and efficient aerodynamic design of turbine cascades. Furthermore, while the current intelligent design framework has been thoroughly validated CFD simulations, advancing it toward practical engineering application necessitates further experimental verification. In future work, the current 2D framework will be extended to 3D turbine blade designs to account for complex spatial flow features. Additionally, detailed linear cascade wind tunnel experiments will be conducted on typical optimized profiles. By acquiring real-world aerodynamic data to further calibrate the prediction model, the practicality and reliability of this intelligent design method will be comprehensively verified for real-world engineering environments.

Data availability

Data will be made available on request.

CRediT authorship contribution statement

Hui Cheng: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Liming Song:** Resources. **Yuqing Ouyang:** Writing – original draft. **Fei Zeng:** Supervision. **Zhendong Guo:** Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was supported by the National Natural Science Foundation of China (Grant No 52576042) and the National Natural Science Foundation of China (Grant No 52306048).

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